Pranshu Tripathi

Software Engineer

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SUMMARY

Software Development Engineer with 2+ years of experience across full-stack development, system design, and AI integrations. Proven expertise in C++, Python, FastAPI, and Kubernetes, with deep knowledge of design patterns, CI/CD, and microservices. Recognized by executive leadership for architecting the LLM-based Maestro Assistant and leading the development team to drive the project from prototype to global launch. Delivered a 300-400x performance boost in molecular calculations using a custom C++ library. Actively seeking challenging SDE roles to build scalable, high-performance systems in cutting-edge tech environments.

EXPERIENCE

Schrödinger

Senior Developer I *January 2026 - Present*

- Project lead of Maestro Assistant.
- Developed a Semantic Panel Search prototype using vector embeddings to replace legacy algorithmic search, successfully validating the architecture for production deployment
- Collaborating on architecture design of Notification and Authentication services for Maestro.

Developer II *July 2024 - December 2025*

- Engineered the "Maestro Assistant" service into a single proxy, implementing CI/CD pipelines and telemetry to support 100+ daily production requests. Received recognition from company leadership and customers like BMS, GSK.
- Implemented non-bonded interaction calculation library; a modular C++ library with caching. It reduced the calculation time by 300-400x; cut 10-min tasks to 30s.
- Mentored an engineering intern on the Project Table Calculator, implementing daily code reviews and enforcing scalable design patterns to ensure the successful delivery of major project components.

Developer I *June 2023 - June 2024*

- Refactored 2000+ lines of legacy code using Factory and Strategy patterns, improving extensibility for handling diverse Job Server outputs.
- Enhanced export functionality for new movie presets (e.g., View Ligand Docking) from Maestro to PyMOL, decreasing manual visualization setup time by 60% and enhancing automated workflow efficiency.

Intern - Software Development *January 2023 - May 2023*

- Created an adapter for Schrödinger's structure class that removed the m2io dependency and enabled a property-writing mechanism with 3x greater flexibility, gaining recognition from Scientific Infrastructure team.
- Implemented a robust Find Toolbar widget following MVC pattern and command pattern with python and C++ inter communication.

Intern - Software Development *May 2022 - July 2022*

- Delivered a robust preset management feature in 2 months—30% faster thanaverage—using Factory and Adapter design patterns.
- \item Decoupled hardcoded knowledge-bot engines into database-driven configuration and added documentation-version inference, enabling per-session documentation corpus selection without code changes or redeployments.
- \item Designed and shipped Maestro Assistant's multi-agent backend framework, including downloader, file-handler, and panel-finder agents with shared tool-execution feedback and reusable agent infrastructure.
- \item Built Maestro Assistant's session and conversation persistence platform with migrated schemas, titled soft-deletable sessions, recency-ordered listings, and mode-scoped REST endpoints.
- \item Closed a critical session-pod trust-boundary gap by removing secrets from container images, pinning dependencies, patching high- and critical-severity CVEs, and tightening session-ID validation.
- \item Delivered an LLM-driven failed-job debugging workflow that diagnoses Maestro job failures and provides computational scientists with actionable explanations instead of raw logs.
- \item Eliminated a user-visible \textasciitilde60-second session-pod startup delay by speculatively warming pods on first keystroke and integrating remaining startup into the first-message flow across new, resumed, and imported sessions.
- \item Led remediation of database connection leaks causing Gunicorn timeouts and backend-wide production sluggishness, restoring stable request latency for Bunsen sessions.
- \item Improved Bunsen file-browser scalability for large directories through server-filtered incremental loading and virtualized rendering, backed by automated performance coverage.
- \item Hardened Bunsen account and session handling by invalidating tokens for pending accounts, securing environment-variable handling, and resolving reconnect loops in paused sessions.
- \item Refactored Bunsen's container handoff and introduced opt-in Auto mode,

EDUCATION

IIIT Allahabad (B.Tech)

Electronics and Communication

Aug 2019 - May 2023

- **GPA: 9.16/10**
- **Batch Rank: 4**
- **Award:** Merit-cum-Means Scholarship for Academic Excellence 2020, 2021, 2022

AWARDS

- Runner-up in Schrödinger Hackathon 2024
- Appreciation from CEO, CTO for Maestro Assistant Project
- Silver Medal - Football, Badminton & Table Tennis at Schrödinger Sports Meet

SKILLS

Programming Languages

- C++, Python, Java, JavaScript, TypeScript, SQL, Shell Scripting, Terraform

Framework & Libraries

- Spring, Express.js, React, Next.js, Qt, Hibernate, PostgreSQL, SIP (Python-C++), PyQt6, BULLJS

Cloud & DevOps

- Docker, Kubernetes (K8s), GitHub Workflows, Travis CI, AWS (Elastic Beanstalk, RDS, ElastiCache), GCP, GKE, G Cloud SQL, NATs Streaming

Generative AI Technologies

- LiteLLM, Vertex AI, Hugging Face, Qdrant Vector DB, LangChain, DSPY, MCP, RAG

Additional Interests

- CNNs, LSTMs, Transformers, Graph DB, IoT Systems, Classification Models, Speech to Text recognition

VOLUNTEERING

- Coordinator - IoT Wing, IIIT Allahabad
- Senior Member - Literature Society, IIIT Allahabad

LANGUAGES

English, Hindi, Marathi, German